Teaching notes

This task can be used when learning about classification and evolutionary trees with GCSE students.

It is based on the article from The Week magazine called 'The abominable snowman's DNA'.

Further information can be found at (links all available at time of publishing):

http://www.theblaze.com/stories/2013/10/17/mystery-solved-dna-test-finds-reasonable-explanation-for-identity-of-abominable-snowman/

Starter

To add an element of mystery to the lesson, show the class a video or an image of a possible abominable snowman, or yeti, sighting:

http://www.youtube.com/watch?v=zZFHxqesLo4
http://www.dailymail.co.uk/news/article-2278024/The-clearest-sighting-Yeti--elaborate-hoax-schoolboys.html

Discuss with the students if they think these sightings are real and what they think a yeti might be.

Then ask them to discuss in pairs what evidence would they need in order to prove that yetis exist. Suitable answers might be samples of fur, footprints, good quality photographs or video.

After hearing their feedback, discuss how their ideas could prove the existence of the yeti and also its limitations. For example, if some fur was found, how could it be proved that it belonged to a yeti?

Main task

Inform students that an Oxford University Professor has collected what could be evidence for the existence of the yeti and their task today is to analyse it.

He has extracted some DNA from two hair samples found in two different mountainous areas where yetis are thought to live and worked out the DNA sequence. He has found they are exactly the same, so from the same species, but are they from the yeti?

Ask students to work in small groups and hand out each group a paper copy of the 'yeti' DNA sequence. Also give them copies of 'human', 'gorilla', 'polar bear' and 'brown bear' DNA sequences which should be photocopied onto acetate. They should follow the tasks given on the worksheet. Students place the acetate copies on top of the 'yeti' sequence and also on top of each other to compare them and help them to answer the questions. The matching sequences are shown at the end of the resource.

Go through what they have found. They should have seen that the polar bear and the brown bear are most closely related (they share seven sequences). The human and gorilla share three sequences. The mystery animal is equally related to both the polar bear and brown bear which suggests that it would be a bear-like creature and not human looking as many 'sightings' of the yeti suggest.
Explain that this is a simulation and in real life the DNA sequences would be much longer (thousands of bases long) which makes the comparison more accurate.

Tell the students that the professor also sequenced DNA from an ancient bear and found a 100% match to the mystery animal - so it looks like the yeti could be a species that was thought to be extinct.

Introduce the idea of common ancestors. They would have seen from the DNA sequencing task that all the animals have one sequence in common which suggests that they all share a common ancestor. Display a mammalian evolutionary tree (http://commons.wikimedia.org/wiki/File:An_evolutionary_tree_of_mammals.jpeg) which shows the orders of Primates (apes belong to this order) and Carnivora (bears belong to this order). Ask the students to place the common ancestor of these two orders on the evolutionary tree.

Display a bear evolutionary tree (http://bix.ucsd.edu/bioalgorithms/downloads/bookfigs/fig.10.7.pdf) and ask the students to place where they think the ancient bear, and therefore the 'yeti', goes.

Plenary

Ask the students to read the article from The Week magazine for further information and then discuss in pairs:

- If this is a species that we thought was extinct, why haven't we discovered it before?
- Can we be sure that this is the 'yeti' that people have claimed to have seen. Why? What further evidence is needed?
- Why is it important that the professor publishes his findings in a scientific journal?

Homework idea

Ask the students to use ideas about natural selection to write an explanation of either:
How the brown and polar bear evolved from this ancient ancestor

or

How the ancient bear may have evolved to become the human-looking 'yeti'

Could the mystery animal be a yeti?

Scientists are able to extract DNA from tissues and sequence it.

Comparing the sequence of bases in a sample of DNA allows them to compare the DNA of organisms.

Closely related organisms will have sequences that are similar.



creative commons image
http://commons.wikimedia.org/wiki/File:Yeti_by_Philippe_Semeria.jpg

Task

You have been given a short piece of DNA from five different sources: a human, gorilla, brown bear, polar bear and the mystery animal that could be a yeti.
Compare the DNA sequences of each animal and:

- State which animals are most closely related and how you know.

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- Suggest which animal/s the mystery animal is most closely related to.

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- Form a hypothesis as to what the mystery animal is.

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Mystery animal 'yeti' DNA

G A C T G G T T C C A A A C T G T G G C C G

C A C T G C A T G C C T A C G T A C G T A A

G T G T G T A A A G T T G C C C G T G C A G

T C C A C A G T G C A A C G T G C A T G C A

T G G G A A C T G G T G T C C A A A T G C A

T G G T G G C A A A C A T C A G A C G T C A

C A G T G T A A C C G T G T G C A A C G T G

C A A C C C G T G G C A G T T G C C C C A A

G T G G C T C A G T G T G C C C A A A G T G

C C G T T G C C G T T G C C A A A C A C G T

G C A G T C A A G A C A C T C T C A A T C G

A T A T A G C G T A C G T G C A A A C C G T

C T A A T G C A T G C A A T T G C C A G T G

C A G T G C T A G T C A A C C T G T G T G C

A T G C A A G T G T T T G T T C A A C G C A

G T T C G T A G T G C A T C T A T G C A T C

Polar bear DNA

A C A C T G T G G C C G T C A G C A T G C A

G G T C A G T C A A C T A C G T A C T G T G

T A A A G T T G C C C G T G C A G C C C A C

A G T G C A A C G T G C A T G C A T G G G T

A C T G G T G C A T T C C A A A T G C A T A

G G T G G C A A A C A T C A G A C G T C A C

A G T G T A A C C G T G T G C A A C G T G C

A A C C C G T G G C A G T T G C C C T A A G

T G G T G T G C C C A A A G T G C G T C A T

C C G T T G C C A A A C A C G T G C T A C T

C T C A A T C G A T G T A G C G T A C G T G

C A A A C C G T G T C A T G C A T G C A A T

T G C C A G T G C A G T T A G T C A A C C T

G T G A G G A T G C A A G T G T T T G C A G

T T C G G A C C G G G A C T T C A A C G A T

C T A T G C A T C G C G G C A T G A C T G G

Brown bear DNA

T G C T A G A A A C T G T G G C C A G C A C

T G C T T G C G T A G T A C G T A G A G A G

A A A G T T G C C C G G T C G A C A G T G C

A A C G T G C A T G C A T G G G A A C T G G

T G T G C T A G T C A C C A A A T G C C T G

G T G G C A A A C A T C A G A C G T C A C A

G T G T A A C C G T G T G C A A C G T G C A

A C G T C G T G G C A G T T G C C C C T T G

T G G C T C A G T G T G C C C A A A G T G C

A C G T T G C C G T C T T C A G A C A C C A

G C G T C G A T G T A G C G T A C A A A C C

G T G T C A T G C A T G C A A T T G C C A G

G T C A G T A A C G A G T A G T C A A C C T

G T G T G C A T G C A A G T G T T T G C A G

T T C G C C A T C A T G T T T C A A C G C C

G A T A C G T C G G C A A G A C T A G C G T

Human DNA

C T T G C T G T G G A C T G T T T G A C G T

A C A T A C A T G C C C G T A A T G A C T C

C C A C A A A G T G C C A A C A G T G A T T

C A T G C A C A T G G G A A A G C C G A T C

A T G C A A G C T G G T C G T C A T G A G A

C G T C A C A G T G T G G T A A C G C G C A

A C G C C T C G C A G A T G C C C C A A G T

G G C T C A G C T G G A C C A A A G T G C G

T C A T A A A A C A C G T G C A G T G T G C

T C G T T G C C A A C A C T C A A T C G A T

G T A G G A T T C G T C C A A A C C G T G T

C A T G G T A C A A A A T G C C A G T G C A

G T G C C A G A C C A G C T G T G T A A A T

C T G A G T A G T T C G T A G T G C A T T C

A G G T A C C G C G G C A T G A C T G G T C

G A G T A G T C C A G T A C T G A C A T G A

Gorilla DNA

T C T G C T G T G T A G A T T G A G C C A G

C A C A T G C C C G T A A G T G T G T A A A

G T T G C C C G T G C A G T C C A C A G T G

C A A C G T G C A T G C A T G G G A A C G T

C C A A A T G C A T G G T G G C A A A C A T

C A G A C G T C A C A G T G T A A C C G T G

T G C A A C C G T G G C A G T T G C C C C A

A G T G G C T C A G T G T G C C C A A A G T

G C G T C A T C A A A C A C G T G C A G T G

A C G T T G C C C T C A A T C G A T G T A G

C G T A C G T G C A A A C C G T G T C A T G

C A T G T G C C A G T G C A G T G C T A A A

C C T G T G T G C A T G C A A G T G T T T G

G T C A A A T G T T C G T A C A G T T C G T

G C A T C T A T G C A T C G C G G C A T G A

C T G G A T G A C G T A C C A G T G C A C T

Solution sheet – mystery animal 'yeti' DNA

G A C T G G T T C C A A A C T G T G G C C G

C A C T G C A T G C C **T A C G T A** C G T A A

G T G T G T A A A G T T G C C C G T G C A G

T C C A C A G T G C A A C G T G C A T G C A

T G G G A A C T G G T G T **C C A A A T G** C A

T G G T G G C A A A C A T C A G A C G T C A

C A G T G T A A C C G T G T G C A A C G T G

C A A C C C **G T G G C A G T T G C C C** C A A

G T G G C T C A G T G T G C C C A A A G T G

C **C G T T G C C** G T T G C C A A A C A C G T

G C A G T C A A G A C A C T C T C A A T C G

A T A T A G C G T A C G T G C A A A C C G T

C T A A T G C A T G C A A T T G C C A G T G

C A G T G C T A G T C A A C C T G T G T G C

A T G C A A G T G T T T G **T T C A A C G** C A

G T T C G T A G T G C A T C T A T G C A T C

Solution key

Colour	Sequence common to
Yellow	all of the animals
Red	brown bear, polar bear and mystery animal
Blue	human and gorilla
Green	polar bear and brown bear

Solution sheet – polar bear DNA

A C A C T G T G G C C G T C A G C A T G C A

G G T C A G T C A A C **T A C G T A** C T G T G

T A A A G T T G C C C G T G C A G C C C A C

A G T G C A A C G T G C A T G C A T G G G T

A C T G G T G C A T T **C C A A A T G** C A T A

G G T G G C A A A C A T C A G A C G T C A C

A G T G T A A C C G T G T G C A A C G T G C

A A C C C **G T G G C A G T T G C C C** T A A G

T G G T G T G C C C A A A G T G C G T C A T

C **C G T T G C C** A A A C A C G T G C T A C T

C T C A A **T C G A T G T A G** C G T A C G T G

C A A A C C G T G T C A T G C A T G C A A T

T G C C A G T G C A G T **T A G T C A A C C T**

G T G A G G A T G C A A G T G T T T G C A G

T T C G G A C C G G G A C **T T C A A C G** A T

C T A T G C A T C G C G G C A T G A C T G G

Solution sheet – brown bear DNA

T G C T A G A A A C T G T G G C C A G C A C

T G C T T G C G T A G **T A C G T A** G A G A G

A A A G T T G C C C G G T C G A C A G T G C

A A C G T G C A T G C A T G G G A A C T G G

T G T G C T A G T C A **C C A A A T G** C C T G

G T G G C A A A C A T C A G A C G T C A C A

G T G T A A C C G T G T G C A A C G T G C A

A C G T C **G T G G C A G T T G C C C C** T T G

T G G C T C A G T G T G C C C A A A G T G C

A **C G T T G C C** G T C T T C A G A C A C C A

G C G **T C G A T G T A G** C G T A C A A A C C

G T G T C A T G C A T G C A A T T G C C A G

G T C A G T A A C G A G **T A G T C A A C C T**

G T G T G C A T G C A A G T G T T T G C A G

T T C G C C A T C A T G T **T T C A A C G** C C

G A T A C G T C G G C A A G A C T A G C G T

Solution sheet – human DNA

C T **T G C T G T G** G A C T G T T T G A C G T

A C A T A C A T G C C C G T A A T G A C T C

C C A C A A A G T G C C A A C A G T G A T T

C A T G C A C A T G G G A A A G C C G A T C

A T G C A A G C T G G T C G T C A T G A G A

C G T C A C A G T G T G G T A A C G C G C A

A C G C C T C G C A G A T G C C C C A A G T

G G C T C A G C T G G A C C A A A G T G C G

T C A T A A A A C A C G T G C A G T G T G C

T **C G T T G C C** A A C A C T C A A T C G A T

G T A G G A T T C G T C C A A A C C G T G T

C A T G G T A C A A A A T G C C A G T G C A

G T G C C A G A C C A G C T G T G T A A A T

C T G A G T A **G T T C G T A** G T G C A T T C

A G G T A C C G C G G C A T G A C T G G T C

G A G T A G T C C A G T A C T G A C A T G A

Solution sheet – gorilla DNA

T C T G C T G T G T A G A T T G A G C C A G

C A C A T G C C C G T A A G T G T G T A A A

G T T G C C C G T G C A G T C C A C A G T G

C A A C G T G C A T G C A T G G G A A C G T

C C A A A T G C A T G G T G G C A A A C A T

C A G A C G T C A C A G T G T A A C C G T G

T G C A A C C G T G G C A G T T G C C C C A

A G T G G C T C A G T G T G C C C A A A G T

G C G T C A T C A A A C A C G T G C A G T G

A C G T T G C C C T C A A T C G A T G T A G

C G T A C G T G C A A A C C G T G T C A T G

C A T G T G C C A G T G C A G T G C T A A A

C C T G T G T G C A T G C A A G T G T T T G

G T C A A A T G T T C G T A C A G T T C G T

G C A T C T A T G C A T C G C G G C A T G A

C T G G A T G A C G T A C C A G T G C A C T